



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.7

Identify Equivalent Algebraic Expressions

Lesson 6-6 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can show that two expressions are equivalent by simplifying and combining like terms.

Language Objective: I can explain my work using the words equivalent, simplify, like terms, and combine.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equivalent Expressions

Equivalent

Simplify

Like Terms

Combine

Coefficient

Commutative Property

Associative Property

Counterexample



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Expressions with the same value for allowed number are equivalent.

2

Expressions that always have the same value are .

3

Rewriting an expression in a shorter form that is still equal to it is to it.

4

Terms with the same variable raised to the same power, like $4x$ and $9x$, are .

5 To add or subtract like terms into one term is to them.

6 A number multiplied by a variable is a .

7 The rule that $a + b = b + a$ is the .

8 The rule that $(a + b) + c = a + (b + c)$ is the .

9 One value that shows two expressions are not equal is a .

Write About the Math

Are these expressions equivalent? How can you prove it?

¿Estas expresiones son equivalentes? ¿Cómo lo puedes probar?

Say your answer to a partner first. Then write it, using at least two of these words:

☐

Equivalent Expressions

Expresiones equivalentes

☐

Equivalent

Equivalente

☐

Simplify

Simplificar

☐

Like Terms

Términos semejantes

☐

Combine

Combinar

Model: $3x + 2x$ combines to $5x$, so the two forms are equivalent. — Students explain $3x + 2x = 5x$ (combining like terms), so both equal $5x + 10$, and verify by substituting a value like $x = 4$ to get 30 both ways.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The expressions ____ equivalent because when I simplify $4h + 2h + 30$, I get ____ . I can prove it by substituting $h =$ ____ : both expressions equal ____ .
- My answer is ____ because ____ .

Mi respuesta es ____ porque ____ .

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____ .
Mi afirmación es ____ .
- My evidence is ____ , so ____ .
Mi evidencia es ____ , entonces ____ .

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Equivalent Expressions
2. **Notes 2:** Equivalent
3. **Notes 3:** Simplify
4. **Notes 4:** Like Terms
5. **Notes 5:** Combine
6. **Notes 6:** Coefficient
7. **Notes 7:** Commutative Property
8. **Notes 8:** Associative Property
9. **Notes 9:** Counterexample
10. **Watch for this mistake:** *A common mistake in Equivalent Expressions is adding the exponents instead of the coefficients when combining like terms — writing $3x + 2x$ as $5x^2$ instead of $5x$. Remember: $3x$ means 3 groups of x and $2x$ means 2 more groups of x , so you add the coefficients ($3 + 2 = 5$) and keep the variable's power the same (x^1 , not x^2).*
11. **Try It 1:** C. $4x - 3x + x$ is 3 groups of x plus 1 more group of $x = 4x$. Add the coefficients: $3 + 1 = 4$. Check $x = 3$: $3(3) + 3 = 12$ and $4(3) = 12$.
12. **Try It 2:** A. $2x + 5$ — Order of addition does not change the value (commutative property): $5 + 2x = 2x + 5$. Check $x = 10$: $5 + 2(10) = 25$ and $2(10) + 5 = 25$.